

A Model-Aware Pipeline for Credit Card Fraud Detection Using Supervised Learning and Anomaly Detection

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Abstract

Credit card fraud detection is a highly imbalanced binary classification problem in which fraudulent activity is rare and costly to miss. This study investigates machine learning methods using the MLG-ULB credit card transaction dataset, comprising 284,807 transactions with 492 labelled fraud cases, equivalent to a fraud rate of approximately 0.17% [1]. The feature space consists of anonymised PCA-transformed variables, together with the raw Time and Amount attributes, limiting direct interpretability while retaining a realistic financial-data setting. A reproducible, leakage safe preprocessing pipeline was developed, covering schema validation, missing value checks, duplicate removal, class distribution analysis, raw feature inspection, and stratified train-test splitting. To support different model assumptions, two data representations were prepared: a minimal unscaled version for tree based and ensemble models, and a robust scaled version for scale sensitive algorithms. The project compares supervised classifiers and anomaly detection methods, including Logistic Regression, Random Forest, XGBoost, LightGBM, Isolation Forest, and One-Class SVM. Given the severe class imbalance, performance is assessed using precision, recall, F1-score, ROC-AUC, and PR-AUC rather than accuracy alone. The aim is to evaluate how different modelling strategies balance fraud detection against false-positive control in a rare-event financial classification task.

1 Introduction

Credit card fraud detection is a rare-event classification problem in which a very small number of fraudulent transactions must be identified within a large volume of legitimate activity, a challenge that remains significant in the wider growth of digital financial fraud [2]. In the dataset used for this project, only 492 out of 284,807 transactions are labelled as fraudulent, giving a fraud ratio of approximately 0.17% [1]. This imbalance makes the task substantially different from standard classification problems, since a model can achieve high apparent accuracy by predicting nearly all transactions as legitimate while failing to detect the minority fraud class.

This project uses the MLG-ULB credit card fraud dataset, sourced from Kaggle. The dataset contains numerical transaction records with 28 anonymised principal component features, denoted as $V1$ to $V28$, together with the raw *Time*, *Amount*, and *Class* variables [1]. The principal component transformation preserves confidentiality but limits the direct interpretation of most predictors. As a result, the modelling task is focused less on explaining individual feature meanings and more on identifying statistical patterns that separate fraudulent and legitimate transactions.

The central challenge addressed in this work is the combination of extreme class imbalance, limited feature interpretability, and the operational trade-off between missed fraud cases and false alarms. Accuracy is therefore not treated as a sufficient measure of performance. Instead, the project evaluates models using metrics that are more informative for imbalanced fraud detection, including precision, recall, F1-score, ROC-AUC, and PR-AUC.

The objective of this project is to develop a reproducible data mining workflow for credit card fraud detection and to compare a range of modelling strategies under a shared preprocessing framework. The methods considered include supervised classifiers, such as Logistic Regression, Random Forest, XGBoost, and LightGBM, as well as anomaly detection approaches, including Isolation Forest and One-Class SVM. The aim is to determine which approaches are most suitable for detecting fraudulent transactions while maintaining an acceptable level of false positives in a highly imbalanced financial transaction setting.

2 Data Preprocessing

The preprocessing stage was designed as a reproducible and leakage-safe pipeline rather than a purely mechanical data cleaning step. Since the dataset is already fully numerical and largely structured, the main objective was to verify data integrity, remove avoidable sources of bias, preserve the severe class imbalance during splitting, and prepare model-specific input representations for later experimentation. The workflow follows a fixed random seed of 42 to ensure that all splits and downstream experiments can be reproduced consistently.

2.1 Raw Data Loading and Schema Validation

The raw dataset was loaded from the project data directory without applying any transformations at import time. This ensured that all subsequent preprocessing decisions were based on the original data structure. The dataset initially contained 284,807 transactions and 31 columns. The expected schema consisted of the raw variables *Time* and *Amount*, the 28 anonymised PCA-transformed variables $V1$ to $V28$, and the binary target variable *Class*[1].

A schema validation step was performed before any cleaning or modelling operation. This check confirmed that all expected columns were present, that the target variable was available, and that the feature set matched the benchmark dataset structure. All feature columns were numeric, with *Time*, *Amount*, and $V1$ to $V28$ stored as floating-point variables, while *Class* was stored as an

integer label. This validation step was included to avoid relying on assumptions about the dataset.

2.2 Basic Data Audit

A column level audit was conducted before transformation, recording data types, missing values, missing value percentages, and unique value counts. The dataset was found to be structurally clean. All variables were numerical, no missing values were present, and the target variable contained only the two expected classes.

Since no imputation, encoding, or type conversion was required, preprocessing focused on methodological preparation for imbalanced classification rather than conventional data cleaning.

2.3 Duplicate Detection and Removal

Exact duplicate rows were identified before the train–test split to prevent repeated observations from influencing training or appearing across both training and test sets. A total of 1,081 exact duplicates were found, representing 0.3796% of the raw dataset. These rows were removed because they contained no additional information and could bias model evaluation.

Table 1: Duplicate removal summary

Item	Value
Rows before duplicate removal	284,807
Exact duplicate rows	1,081
Duplicate percentage	0.3796%
Rows after duplicate removal	283,726

2.4 Class Imbalance Analysis

After duplicate removal, the distribution of the target variable was analysed. The dataset remained extremely imbalanced, with 283,253 legitimate transactions and only 473 fraudulent transactions. This corresponds to a fraud proportion of 0.1667%.

Table 2: Class distribution after duplicate removal

Class	Count	Percentage
Legitimate transactions	283,253	99.8333%
Fraudulent transactions	473	0.1667%

This imbalance is central to the design of the project. A classifier that predicts every transaction as legitimate would obtain very high accuracy while detecting no fraud cases. For this reason, accuracy was not treated as a sufficient evaluation metric. The preprocessing stage therefore preserved the natural imbalance for evaluation and prepared the data so that later models could be assessed using precision, recall, F1-score, ROC-AUC, and PR-AUC.

2.5 Inspection of Raw Numerical Features

Although most predictors are PCA-transformed and anonymised, the dataset also includes two original numerical variables: *Time* and *Amount*. These variables were inspected separately because

they may have different distributions and scales compared with the PCA features.

The *Time* variable represents elapsed time between transactions and ranged from 0 to 172,792 seconds. Its skewness was approximately -0.036, indicating no substantial skew. In contrast, *Amount* was strongly right-skewed, with a skewness of approximately 16.979. Most transactions had relatively small values, while a small number of transactions had very large amounts. The maximum transaction amount was 25,691.16, while the median was only 22.00.

Table 3: Summary statistics for raw numerical features

Statistic	<i>Time</i>	<i>Amount</i>
Mean	94,811.08	88.47
Standard deviation	47,481.05	250.40
Median	84,692.50	22.00
Maximum	172,792.00	25,691.16
Skewness	-0.036	16.979

This inspection directly informed the scaling strategy. The strong skewness in *Amount*, together with the presence of rare and high-value transactions, suggested that scale-sensitive models would benefit from a robust scaling approach rather than standardisation based on mean and variance alone.

2.6 Feature Grouping

Before splitting the data, the variables were explicitly grouped into target, raw numerical features, and PCA-transformed features. The target column was defined as *Class*. The raw feature group contained *Time* and *Amount*. The PCA feature group contained *V1* to *V28*. The final feature matrix therefore contained 30 predictors.

This grouping made the workflow clearer and reduced the risk of inconsistent feature selection across later modelling notebooks. It also allowed the same feature definitions to be reused when creating both the minimal and robust-scaled preprocessing pathways.

2.7 Leakage-Safe Train-Test Split

The train-test split was performed before fitting any preprocessing transformation. This was necessary to prevent data leakage, particularly during scaling. If a scaler were fitted before splitting, information from the test distribution would influence the transformed training data and could lead to overly optimistic evaluation results.

An 80:20 stratified split was used with the fixed random seed. Stratification was essential because the fraud class is extremely rare. Without stratification, the test set could contain an unstable or unrepresentative number of fraud cases.

Table 4: Stratified train–test split.

Subset	Total Transactions	Legitimate	Fraud
Training Set	226,980	226,602	378
Test Set	56,746	56,651	95

The resulting class proportions were preserved across both subsets. Fraud represented 0.1665% of the training set and 0.1674% of the test set. This confirmed that the split retained the original imbalance structure and provided a realistic evaluation set.

2.8 Cross-Validation Configuration

A stratified five-fold cross-validation strategy was also defined for later modelling. The configuration used `StratifiedKFold` with five splits, shuffling enabled, and the same random seed of 42. Cross-validation was not executed during preprocessing; instead, the configuration was established so that later modelling experiments could use a consistent evaluation design.

Stratified cross-validation was selected instead of ordinary K-fold cross-validation because the fraud class is rare. Without stratification, some folds could contain too few fraudulent transactions, making performance estimates unstable. Five folds provided a practical balance between evaluation reliability and computational cost.

2.9 Model-Aware Preprocessing Design

A single preprocessing representation was not imposed on all models because the project compares algorithms with different assumptions. Tree-based and ensemble models such as Random Forest, XGBoost, and LightGBM are generally less sensitive to feature magnitude because they rely on split-based decision rules. Isolation Forest was also treated as suitable for the minimal pathway because its anomaly detection mechanism does not require the same scale assumptions as linear or kernel-based methods.

In contrast, Logistic Regression and One-Class SVM are more sensitive to feature scale. For these models, differences in feature magnitude can affect optimisation, distance-based behaviour, or decision boundaries. Therefore, two parallel preprocessing pathways were created.

This design allowed each model family to use an appropriate representation while still preserving a shared split and target structure across the group project.

2.10 Minimal Preprocessing Pathway

The minimal pathway preserved the original feature values after duplicate removal and stratified splitting. No scaling, sampling, or feature transformation was applied. This representation was used for models that are less sensitive to feature magnitude, including Random Forest, XGBoost, LightGBM, and Isolation Forest.

The resulting matrices contained 226,980 training observations and 56,746 test observations, each with 30 features.

2.11 Robust-Scaled Preprocessing Pathway

For scale-sensitive models, a robust-scaled pathway was created using `RobustScaler`. This method was selected because *Amount* was strongly right-skewed, with a skewness of 16.979, making median and interquartile-range scaling more appropriate than mean-variance standardisation.

To avoid leakage, the scaler was fitted only on the training set and then applied to both training and test features. The scaled matrices retained the same dimensions as the minimal pathway: 226,980 training observations and 56,746 test observations, each with 30 features. [9]

2.12 Sampling Decision

No resampling was applied during preprocessing. This preserved the natural fraud prevalence in both the training and test sets and avoided introducing leakage before model evaluation.

Any later use of undersampling or SMOTE should be treated as a model-training decision and applied only within the training workflow, ideally inside cross-validation folds. The test set was deliberately left unchanged to reflect the real class distribution. [8, 10]

2.13 Exported Outputs and Metadata

The final preprocessing outputs were exported as separate files to support reuse across modelling notebooks. The exported files included the minimal training and test matrices, the robust-scaled training and test matrices, and the corresponding training and test labels.

A metadata file was also saved to record the key preprocessing decisions. This file stored the random seed, original row count, final row count, number of duplicates removed, split ratio, cross-validation configuration, scaling method, target column, feature groups, and notes describing the main methodological choices.

Table 5: Exported preprocessing outputs

Output file	Purpose
<code>X_train_min.csv</code>	Unscaled training features
<code>X_test_min.csv</code>	Unscaled test features
<code>X_train_rs.csv</code>	Robust-scaled training features
<code>X_test_rs.csv</code>	Robust-scaled test features
<code>y_train.csv</code>	Training labels
<code>y_test.csv</code>	Test labels
<code>pre_metadata.json</code>	Reproducibility metadata

3 Methods and Experimentation

The modelling stage compared supervised and anomaly-based approaches under the shared leakage-safe preprocessing framework. Given the fraud rate of only 0.17%, experiments were not framed as accuracy maximisation; models were assessed by their ability to recover rare fraud cases while limiting false positives among legitimate transactions.

Supervised classifiers, including Logistic Regression, LightGBM, XGBoost, and Random Forest, were evaluated alongside anomaly detection methods, including Isolation Forest and One-Class SVM. This comparison tested whether fraudulent transactions are better captured through labelled decision boundaries or as anomalous behaviour within the wider transaction distribution.

All models used the same stratified train-test split. Tree-based and ensemble models used the minimal unscaled data, while scale-sensitive models used the robust-scaled representation. Performance was evaluated using precision, recall, F1-score, ROC-AUC, and PR-AUC, with an emphasis on recall and PR-AUC due to the extreme class imbalance.

3.1 Baseline and Comparative Models

Logistic Regression was used as a linear supervised baseline. The standard version produced a conservative classifier with high fraud precision but lower recall, detecting 55 out of 95 fraud cases in the test set. A class-weighted version increased recall substantially by detecting 83 fraud cases, but this came at the cost of a large number of false positives and a sharp reduction in fraud-class precision. This result illustrates the main imbalance trade-off, increasing sensitivity to fraud can rapidly reduce operational usefulness if too many legitimate transactions are incorrectly flagged.

Anomaly detection methods provided a contrasting modelling perspective. One-Class SVM detected 36 out of 95 fraud cases and achieved a PR-AUC of 0.2364, which is substantially above the random baseline implied by the fraud rate, but still weaker than the strongest supervised models. Isolation Forest was tested using multiple configurations because its behaviour is highly dependent on the contamination setting. The baseline configuration detected more fraud cases but generated many false positives, while contamination-aligned and tuned configurations reduced false alarms at the cost of missing more fraud cases. The tuned Isolation Forest configuration detected 35 fraud cases with 214 false positives, giving a more balanced but still limited anomaly-detection result.

LightGBM provided one of the strongest supervised results. Using class-imbalance weighting through `scale_pos_weight`, it detected 73 out of 95 fraud cases while incorrectly flagging only 11 legitimate transactions. This produced a strong fraud-class precision of 0.87, recall of 0.77, F1-score of 0.82, ROC-AUC of 0.9602, and PR-AUC of 0.804. These results show that gradient boosting was able to model the minority class effectively without producing excessive false positives.[5]

XGBoost was also evaluated as a gradient-boosted tree model, with two imbalance-handling strategies: Class weighting through `scale_pos_weight` and Synthetic minority oversampling using SMOTE.

The class-weighted XGBoost model used `scale_pos_weight = 577` and achieved strong discrimination, with ROC-AUC of 0.9800 and AUPRC of 0.8796. It detected 83 out of 98 fraud cases with fraud precision of 0.874, while producing only 12 false positives. The SMOTE-based XGBoost variant achieved slightly higher recall of 0.867, but its precision dropped to 0.616 and the false-positive count increased substantially to 53. This showed that, for XGBoost, class weighting provided a more stable precision-recall trade-off than oversampling.[6]

Random Forest was subsequently evaluated as a leading candidate due to its ability to model non-linear feature interactions without requiring feature scaling. The initial configuration detected 68 fraudulent transactions but produced 3,332 false positives, indicating high sensitivity with weak precision control. Applying undersampling improved minority-class recall to 0.86 and reduced false positives to 943. However, the false-positive burden remained substantial. Despite this limitation, the undersampled Random Forest delivered the strongest preliminary fraud-detection performance and was therefore selected as the primary model family for further fine-tuning.

3.2 Model Selection and Fine-Tuning Direction

The comparative results show that the highest recall models were not necessarily the most suitable for fraud detection in practice. Class-weighted Logistic Regression and undersampled Random Forest recovered more fraud cases, but both generated a large number of false positives. XGBoost demonstrated that class weighting can be highly effective for imbalanced fraud detection, achieving strong AUPRC and precision while maintaining high recall. However, the SMOTE-based XGBoost variant also showed that oversampling can increase recall at the expense of false-positive control.

Across the preliminary experiments, tree-based ensemble methods provided the most reliable performance under extreme imbalance. Random Forest was selected as the final model family for further refinement because it offered a strong balance between minority-class recall, non-linear feature modelling, and compatibility with resampling-based training strategies. The next stage therefore focused on fine-tuning the Random Forest-SMOTE configuration to improve fraud detection while reducing the false-positive burden observed in earlier Random Forest variants.

3.3 Random Forest

Having established a linear baseline with Logistic Regression, a Random Forest classifier was evaluated to capture potential non-linear interactions within the high-dimensional, PCA-transformed feature space (V1–V28). Because its bagging architecture reduces variance by constructing an ensemble of decision trees on bootstrapped subsets, Random Forest is naturally more robust to the sparse and irregular patterns typical of fraud data.

To determine the most effective approach for handling the extreme class imbalance, three configurations were systematically tested. A class-weighted baseline on the original data, a randomly undersampled model, and a SMOTE-augmented model. While the baseline and undersampled models succeeded in capturing fraud, they introduced severe operational bottlenecks, generating 3,332 and 943 false positives respectively. Because discarding majority-class data or relying solely on algorithmic weighting proved insufficient for precise boundary delineation, we moved toward synthetic over-sampling to preserve the full distribution of legitimate transactions.

The subsequent sections detail these experiments, evaluated against the same stratified test set ($n = 56,746$ containing 95 fraud cases), leading to the selection of the SMOTE variant for final fine-tuning.

3.3.1 Random Forest using SMOTE. While the baseline and undersampled Random Forest configurations demonstrated high sensitivity, both were ultimately hindered by severe precision deficits. The baseline model generated an unmanageable 3,332 false positives, rendering it operationally unviable. Undersampling attempted to correct this and achieved a strong recall of 0.86, but it still produced 943 false alarms. This high false-positive burden is a direct consequence of discarding critical majority-class information, which prevents the model from accurately mapping legitimate transaction behaviour. To resolve these limitations, a Synthetic Minority Oversampling Technique (SMOTE) was applied. Rather than discarding data, SMOTE preserved all legitimate transactions while synthetically augmenting the fraud class via feature-space interpolation.

This yielded a fully balanced training set of 453,204 observations. [8]

A more expressive Random Forest (max_depth=26, max_features=0.5, n_estimators=100) was then trained on this augmented dataset. This configuration delivered the strongest performance across the evaluation. Fraud precision improved dramatically to 0.92, meaning 92% of flagged transactions were genuine, a qualitative shift from the near-zero precision of the earlier models. Although recall dropped slightly to 0.76, reflecting the classical precision-recall trade-off, the model's overall reliability improved substantially because it became significantly more selective.

By retaining full visibility of the legitimate transaction space, the model established tighter discriminative boundaries, achieving a leading PR-AUC of 0.814 and an F1-score of 0.83. While SMOTE carries the theoretical risk of generating synthetic points in ambiguous regions of the feature space, the empirical results confirm that it generalises significantly better than random undersampling on this dataset. Consequently, the SMOTE-trained Random Forest was selected as the definitive foundational model to carry forward for fine-tuning.

4 Results and Discussion

The experimental results demonstrate that performance on this dataset cannot be interpreted through fraud detection counts alone. Several models achieved high recall by increasing their sensitivity to the minority class, but this often resulted in a large number of false positives. This trade-off is particularly important in credit card fraud detection, where excessive false alarms can reduce operational efficiency and undermine the practical value of the model.[4]

The linear baseline established this issue clearly. Standard Logistic Regression achieved strong precision of 0.85 but detected only 55 of the 95 fraud cases in the test set. Introducing class weighting increased recall to 0.87 by detecting 83 fraud cases, but precision dropped sharply to 0.06 due to 1,389 false positives. This confirmed that algorithmic imbalance correction alone can improve sensitivity but may produce an impractical decision boundary under severe class imbalance.[10]

The anomaly detection models were less competitive than the strongest supervised approaches. One-Class SVM and the tuned Isolation Forest both detected fewer than 40 fraud cases and produced lower F1-scores. Their performance suggests that, although

fraud can be partially represented as anomalous behaviour, the labelled supervised setting provides substantially stronger discriminative information for this dataset. This is consistent with the PCA-transformed feature space, where fraudulent and legitimate transactions may not be separable purely as outliers.

Gradient boosting methods produced stronger results. LightGBM achieved a precision of 0.87, recall of 0.77, and PR-AUC of 0.804, demonstrating an effective balance between minority-class detection and false-positive control. XGBoost was the strongest comparative benchmark, particularly when class weighting was used. The class-weighted XGBoost model achieved an AUPRC of 0.8796 and detected 83 fraud cases while producing only 12 false positives. In contrast, the SMOTE-based XGBoost variant slightly improved recall but reduced precision, showing that oversampling did not improve the overall precision-recall balance for this model family.

Random Forest provided the most suitable foundation for the final model development because it responded effectively to resampling while preserving strong non-linear modelling capacity. The baseline and undersampled configurations showed that the model could identify fraudulent cases, but their high false-positive counts limited practical usefulness. The SMOTE-trained Random Forest offered a more stable balance by preserving the full legitimate transaction distribution while synthetically augmenting the minority class. This produced a substantially more selective classifier, with fraud precision of 0.92, recall of 0.76, F1-score of 0.83, and PR-AUC of 0.814.

Overall, the results show that the best model is not simply the one with the highest recall or the highest ROC-AUC. In this rare-event setting, PR-AUC and fraud-class precision provide a more realistic view of model quality because they reflect performance on the minority class directly. The final Random Forest-SMOTE model was therefore selected not because it maximised every metric, but because it provided the most defensible operational trade-off. It recovered a substantial proportion of fraud cases while sharply reducing the false-positive burden observed in earlier Random Forest configurations.[7]

Table 6: Overall comparative performance of all evaluated models on the held-out test set

Model	Fraud Detected	False Positives	Precision	Recall	F1-score	PR-AUC
Logistic Regression	55	10	0.85	0.58	0.69	0.6980
Class-weighted Logistic Regression	83	1389	0.06	0.87	0.11	0.6720
One-Class SVM	36	166	0.18	0.38	0.24	0.2364
Isolation Forest (tuned)	35	214	0.14	0.37	0.20	0.6823
LightGBM	73	11	0.87	0.77	0.82	0.8040
XGBoost (class-weighted)	83	12	0.874	0.847	–	0.8796
XGBoost (SMOTE)	85	53	0.616	0.867	–	0.8675
Random Forest (original)	68	3332	0.99	0.72	0.83	0.2470
Random Forest (undersampled)	82	943	0.08	0.86	0.15	0.7020
Random Forest (SMOTE)	72	6	0.92	0.76	0.83	0.8140

5 Conclusion

This project developed a model-aware machine learning pipeline for rare-event credit card fraud detection using the MLG-ULB transaction dataset. The main challenge was the extreme class imbalance, with fraudulent transactions representing only approximately 0.17% of the dataset. To address this, the project used a leakage-safe pre-processing workflow, preserved a stratified held-out test set, and evaluated models using imbalance-sensitive metrics rather than accuracy alone.

The experiments showed that supervised learning methods were more effective than unsupervised anomaly detection for this dataset. Logistic Regression provided a useful baseline but was highly sensitive to the precision-recall trade-off when class weighting was applied. One-Class SVM and Isolation Forest detected some fraudulent behaviour but were limited by lower recall and weaker precision. Gradient boosting methods, particularly XGBoost and LightGBM, achieved strong performance and demonstrated the effectiveness of class weighting under severe imbalance.

Random Forest was selected as the final modelling direction because it combined non-linear feature modelling with strong compatibility with resampling-based imbalance handling. The SMOTE-trained Random Forest improved precision control compared with the earlier baseline and undersampled configurations, while still detecting a substantial proportion of fraudulent transactions. This made it the most suitable final model for balancing fraud recovery against false-positive reduction.

The results highlight that credit card fraud detection should not be treated as a conventional classification task. Under extreme imbalance, high accuracy is largely uninformative, and model selection must instead focus on minority-class recall, fraud precision, F1-score, and PR-AUC. Future work could extend this pipeline by performing threshold optimisation, cost-sensitive evaluation, and further hyperparameter tuning to align model decisions more closely with real-world fraud investigation costs. [3]

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